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Using Lattice Boltzmann to Inform Graph Diffusion Networks for Aerodynamic Design Exploration

DSFD 2026

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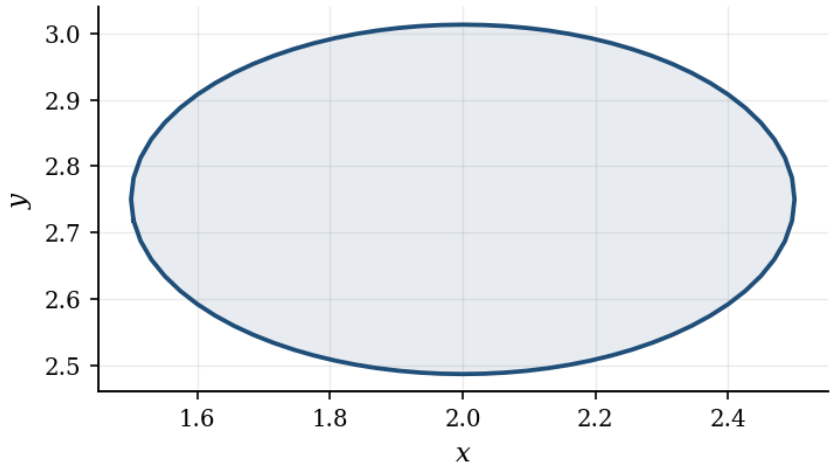
Department of Mechanical and Aerospace Engineering | The University of Manchester

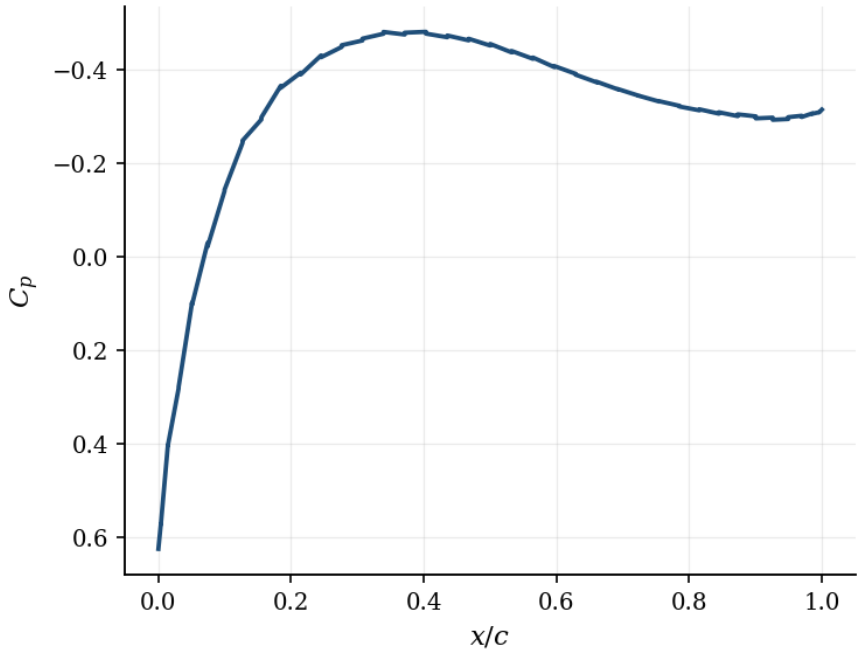
Email: archie.dobson@postgrad.manchester.ac.uk

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Problem Formulation

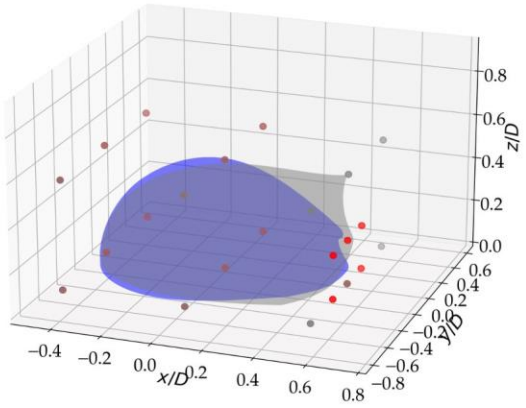
Generate shapes consistent with a target surface pressure utilising Graph Diffusion Network operating directly on the boundary





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GDN for Aerodynamics and utility of LBM (OpenLB)



1. Graph Network
   - Operation Directly on Geometry Nodes
   - Non-Parametric Optimisation

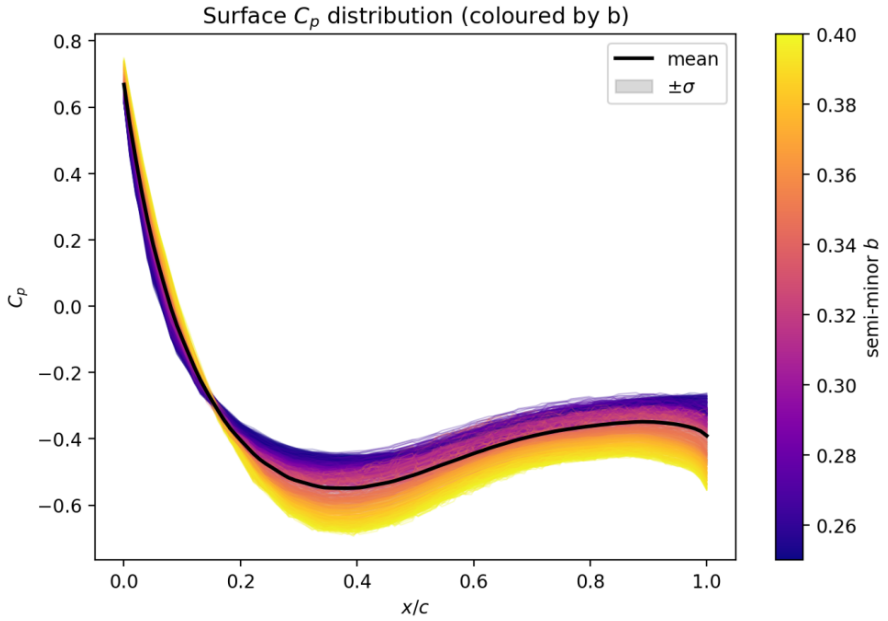
2. LBM as the "ground truth"
   - Ability to handle complex geometries
   - Minimal/no pre or post processing
   - Structural Similarities to GNN – Streaming and Collision

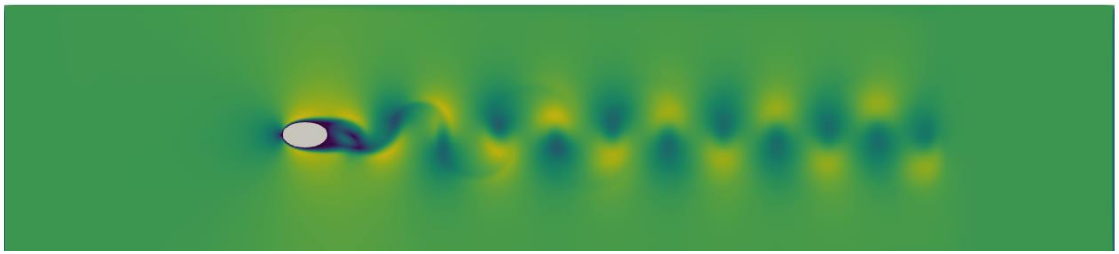
[1] Krause, M. J. et al. (2021). OpenLB - Open source lattice Boltzmann code. Computers & Mathematics with Applications, 81, 258–288. https://doi.org/10.1016/j.camwa.2020.04.033

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Dataset and LBM Validation

- Re 500-1000
- Set of C_p produced





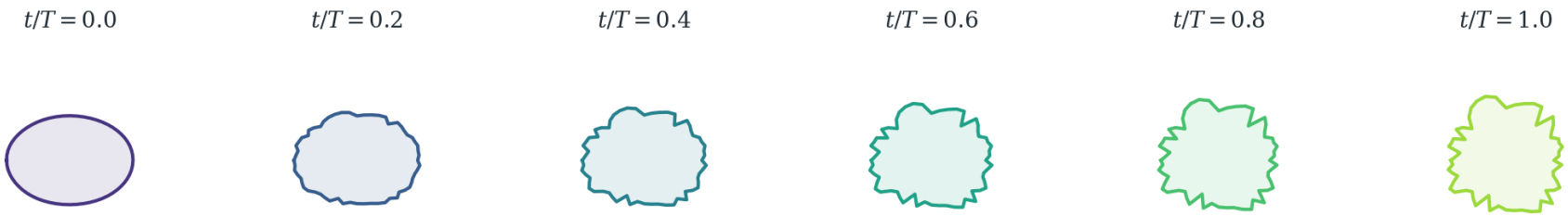


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Overview of Diffusion Models

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}\right)$$

forward $q(x_t \mid x_0)$: add noise ⟶



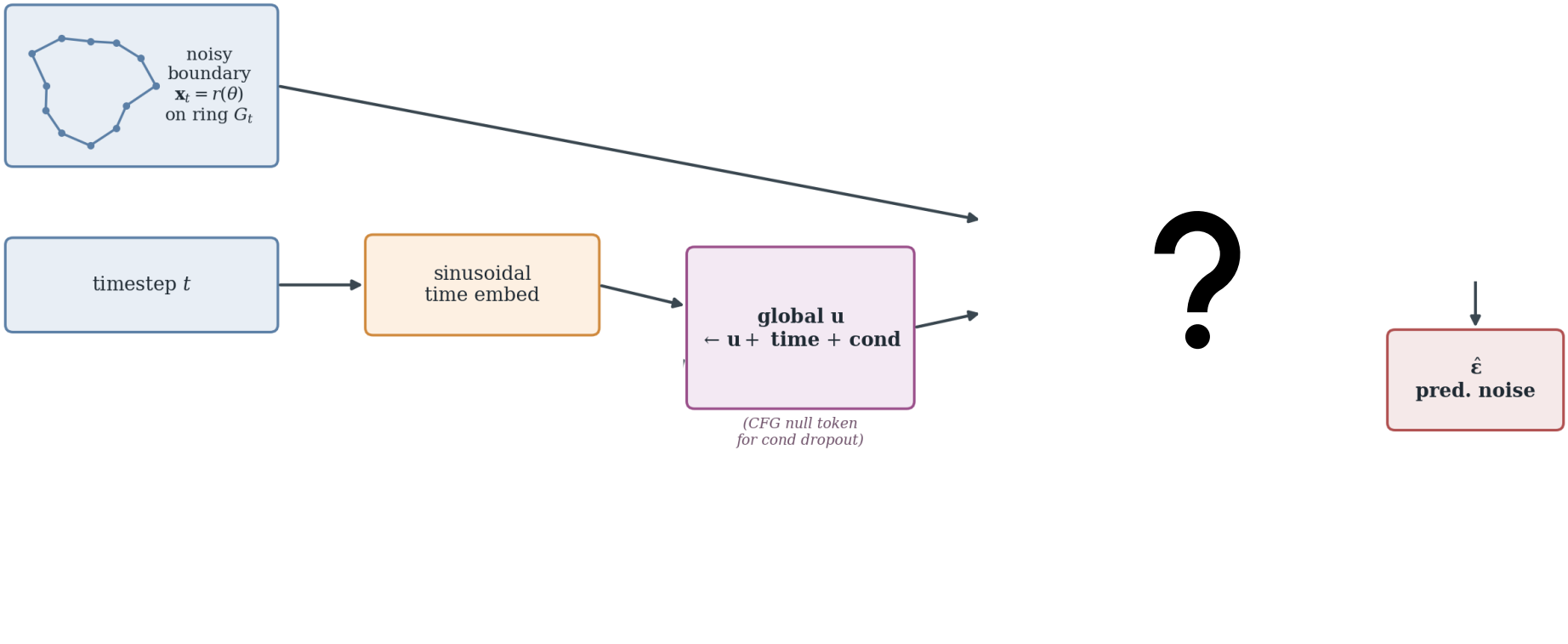
$$p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{x}_t, t),\ \boldsymbol{\Sigma}_\theta)\ .$$

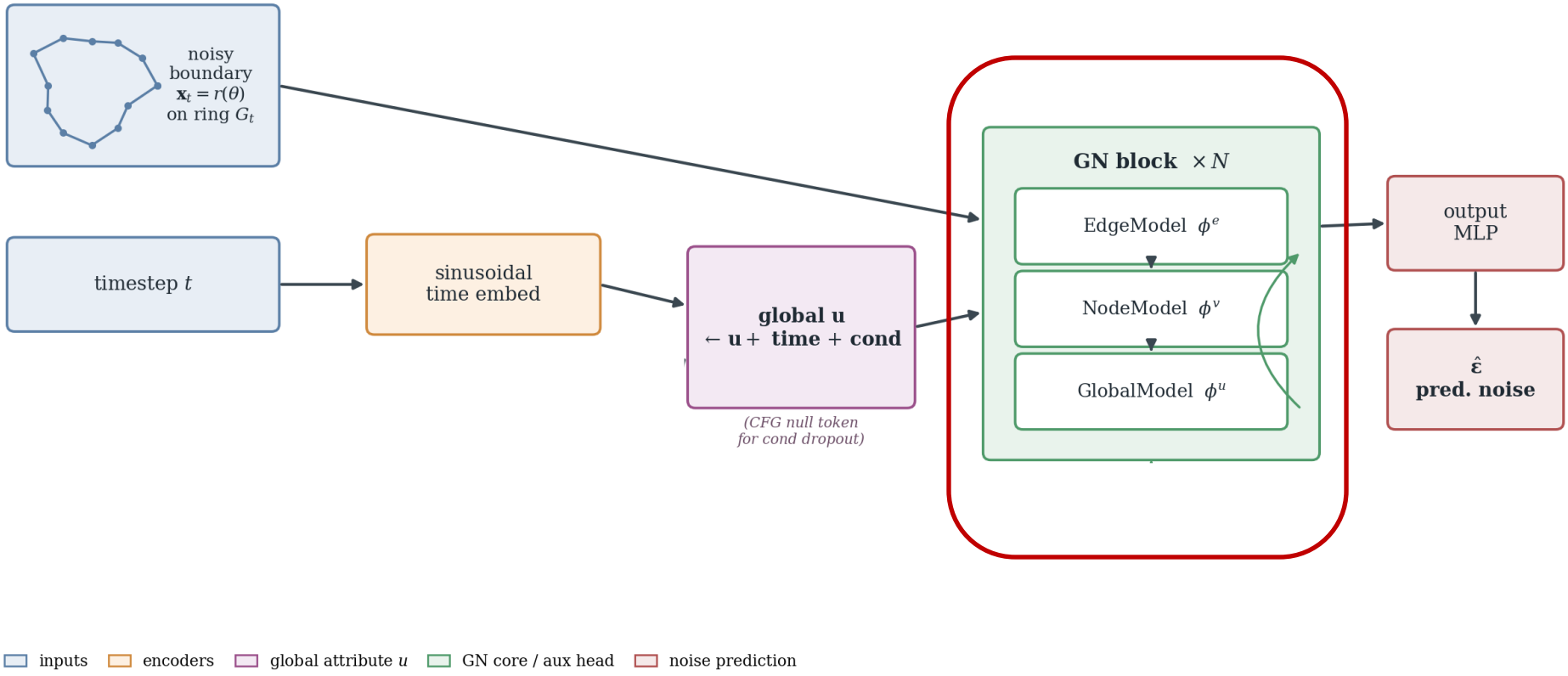
⟵ reverse $p_\theta(x_{t-1} \mid x_t)$: denoise

θ terms = estimated quantities

[1] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. Advances in Neural Information Processing Systems, 33, 6840–6851. https://doi.org/10.48550/arxiv.2006.11239
[2] Song, J., Meng, C., & Ermon, S. (2020). Denoising diffusion implicit models. arXiv preprint arXiv:2010.02502. https://doi.org/10.48550/arXiv.2010.02502

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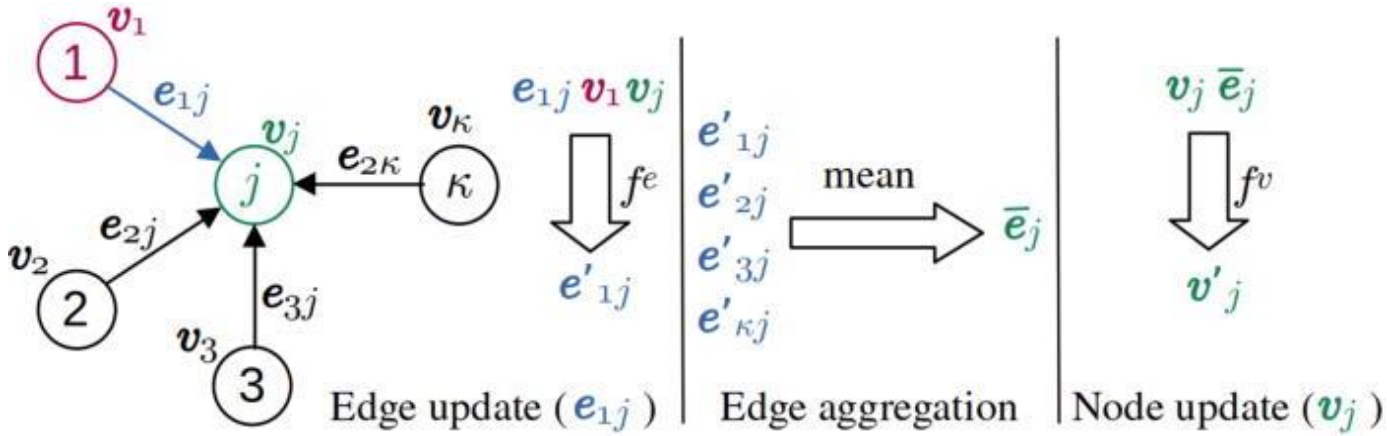

noisy boundary $\mathbf{x}_t = r(\theta)$ on ring G_t
timestep t
sinusoidal time embed
global u ← u + time + cond
(CFG null token for cond dropout)
GN block ×N
EdgeModel ϕ^e
NodeModel ϕ^v
GlobalModel ϕ^u
output MLP
$\hat{\boldsymbol{\varepsilon}}$ pred. noise
inputs
encoders
global attribute u
GN core / aux head
noise prediction

Score Model MuS-GNN --- AIRTUK Tutorial

- Traditionally CNN



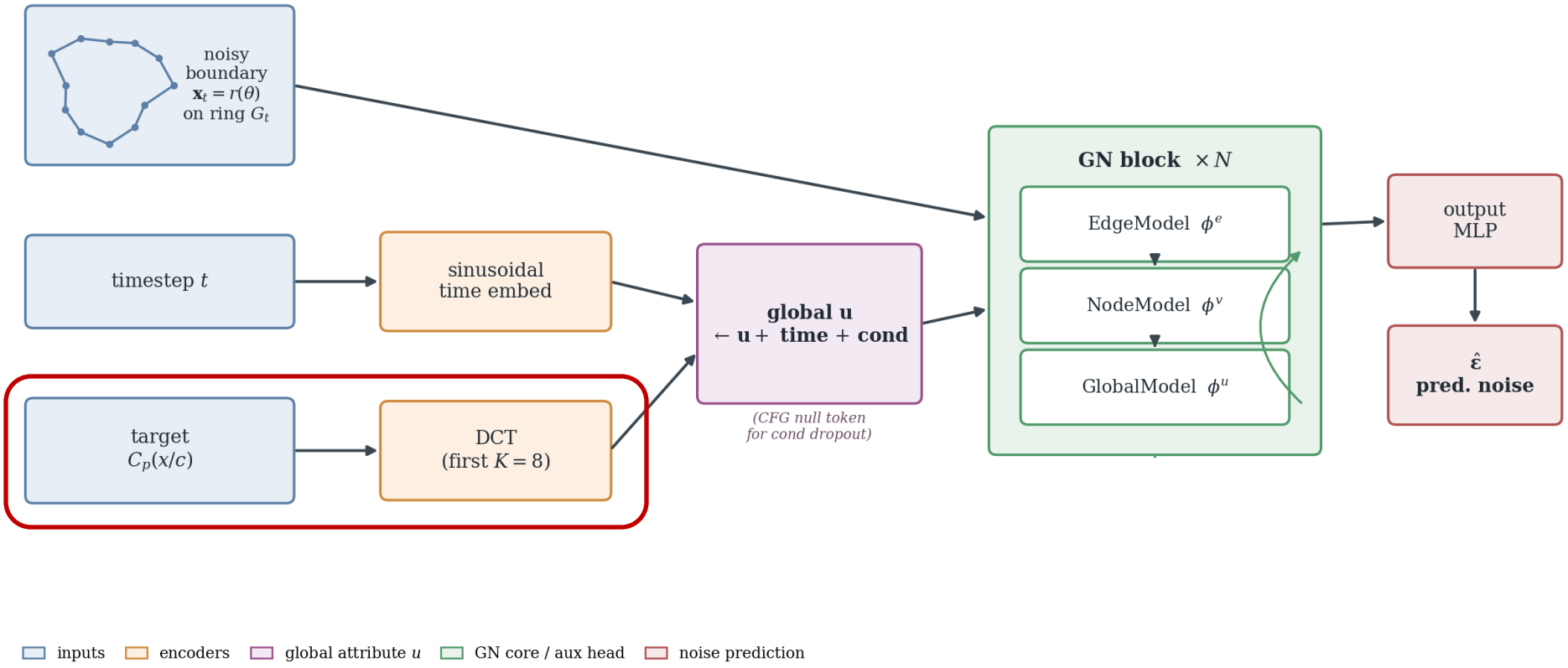
- Graph representation allows direct geometry operation



[1] Battaglia, P. W., et al. (2018). Relational inductive biases, deep learning, and graph networks. arXiv preprint arXiv:1806.01261. https://doi.org/10.48550/arXiv.1806.01261

[2] Lino, M., et al. (2022). Multi-scale rotation-equivariant graph neural networks for unsteady Eulerian fluid dynamics. Physics of Fluids, 34(8), 087110. https://doi.org/10.1063/5.0097679

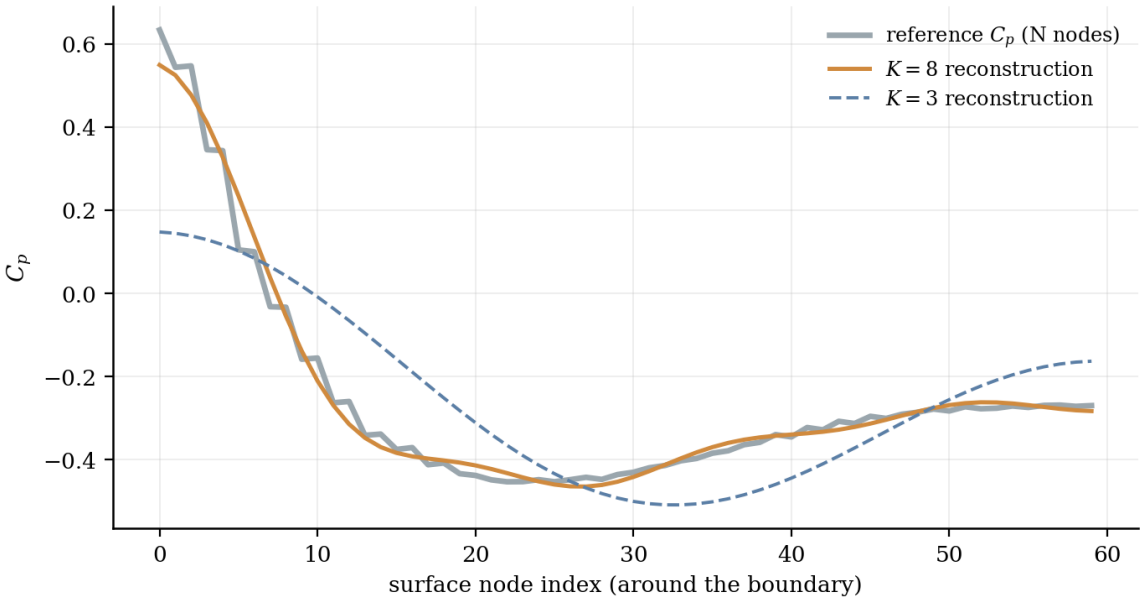
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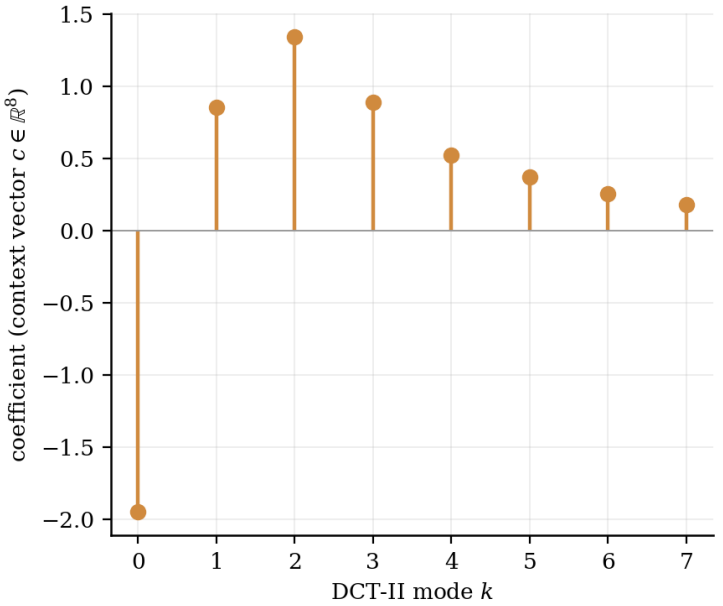


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Ellipses with Pressure Conditioning

- A discrete cosine transform (DCT) keeps the first 8 coefficients.
- Result: a summary of the target, independent of number of surface nodes.

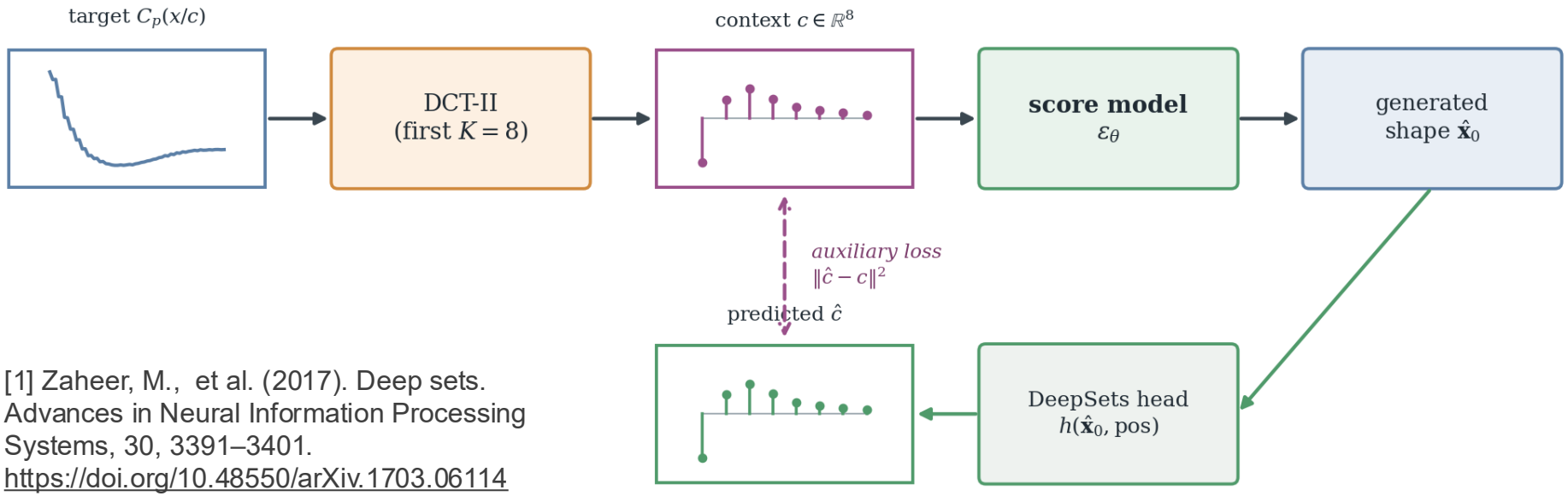




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Ellipses with Pressure Conditioning Cont.

- The target is injected into the **score model.**
- A **DeepSets[1] head reads the generated shape back** and re-predicts its pressure code ĉ.
- An **auxiliary loss** $\|\hat{c} - c\|^2$ reinforcing conditioning.

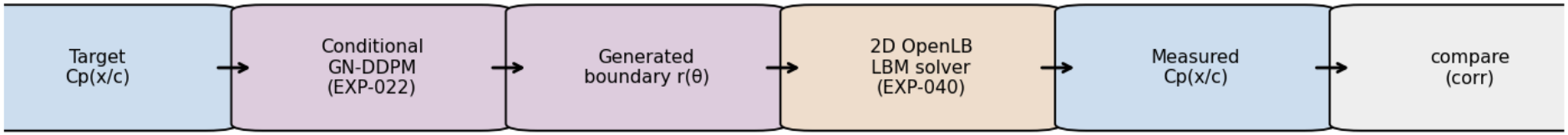


[1] Zaheer, M., et al. (2017). Deep sets. Advances in Neural Information Processing Systems, 30, 3391–3401. https://doi.org/10.48550/arXiv.1703.06114

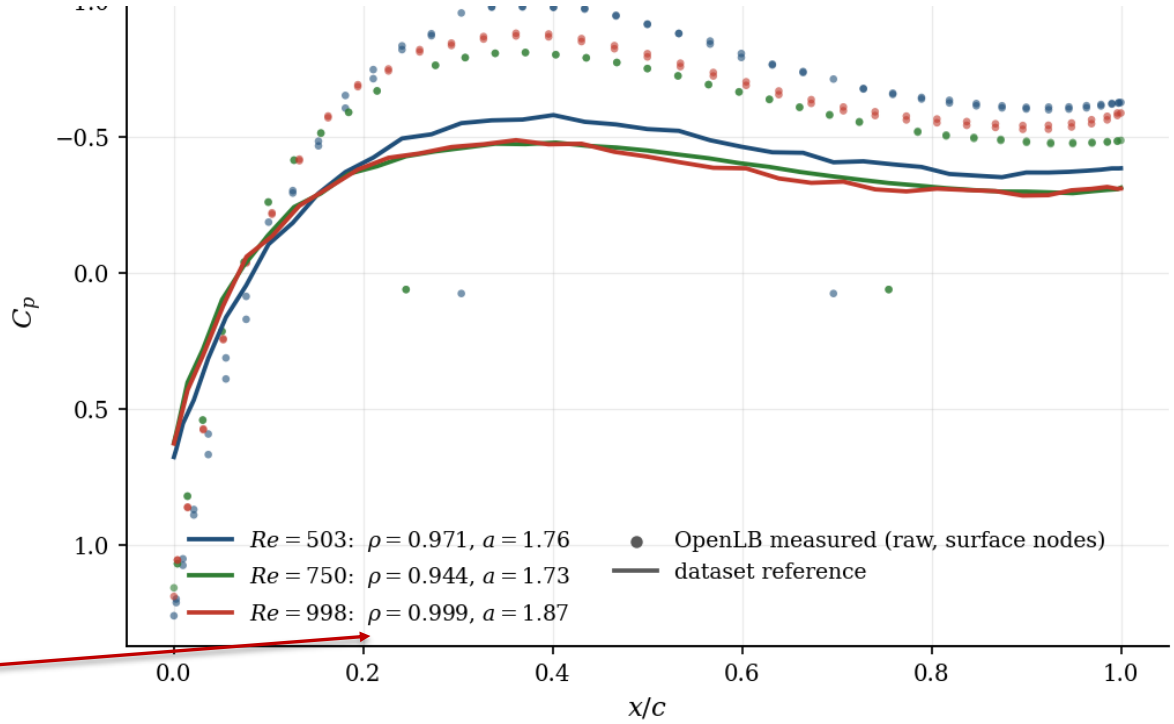
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$$\rho = \frac{\mathrm{cov}(x,y)}{\sigma_x, \sigma_y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}, \sqrt{\sum_i (y_i - \bar{y})^2}}$$

Conditioned Generation



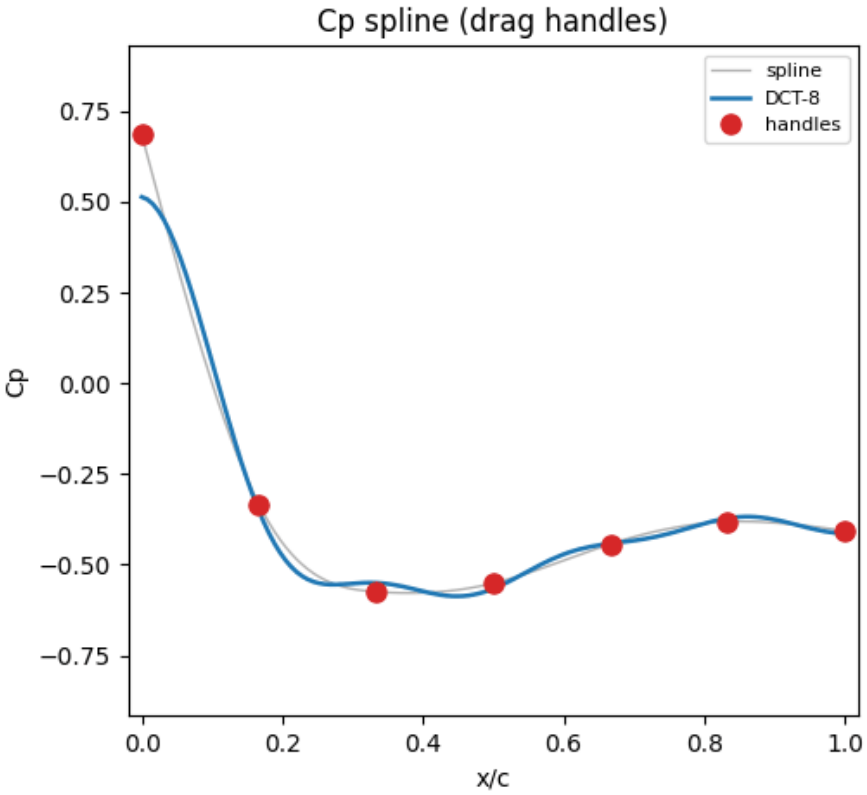
- Pressure-conditioned generation reproduces target C_p trends with high in-distribution correlation
- Systematic error leading to multiplicative scaling

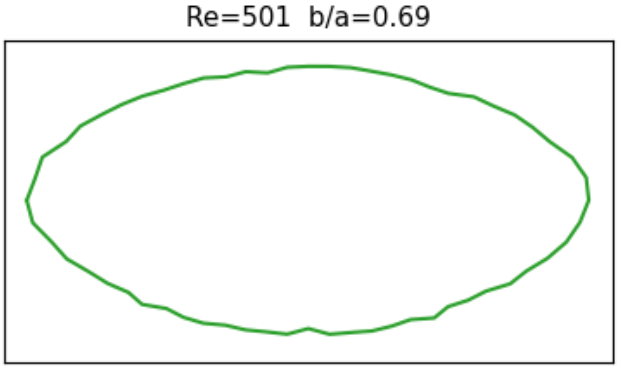


Pearson Correlation

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Exploring the Design Space



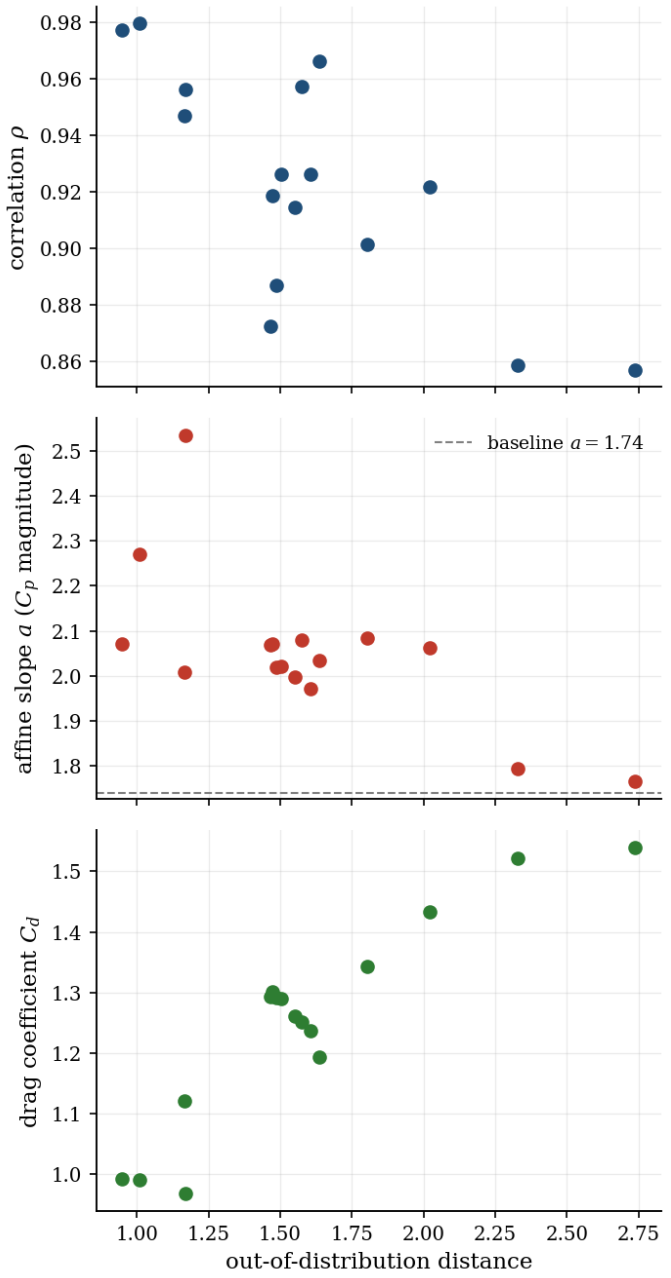


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Limitations

- As out-of-distribution distance grows:
  - Correlation drops
  - Pressure magnitude increase
  - Drag drifts up

- So the method captures near the data, speculative beyond it



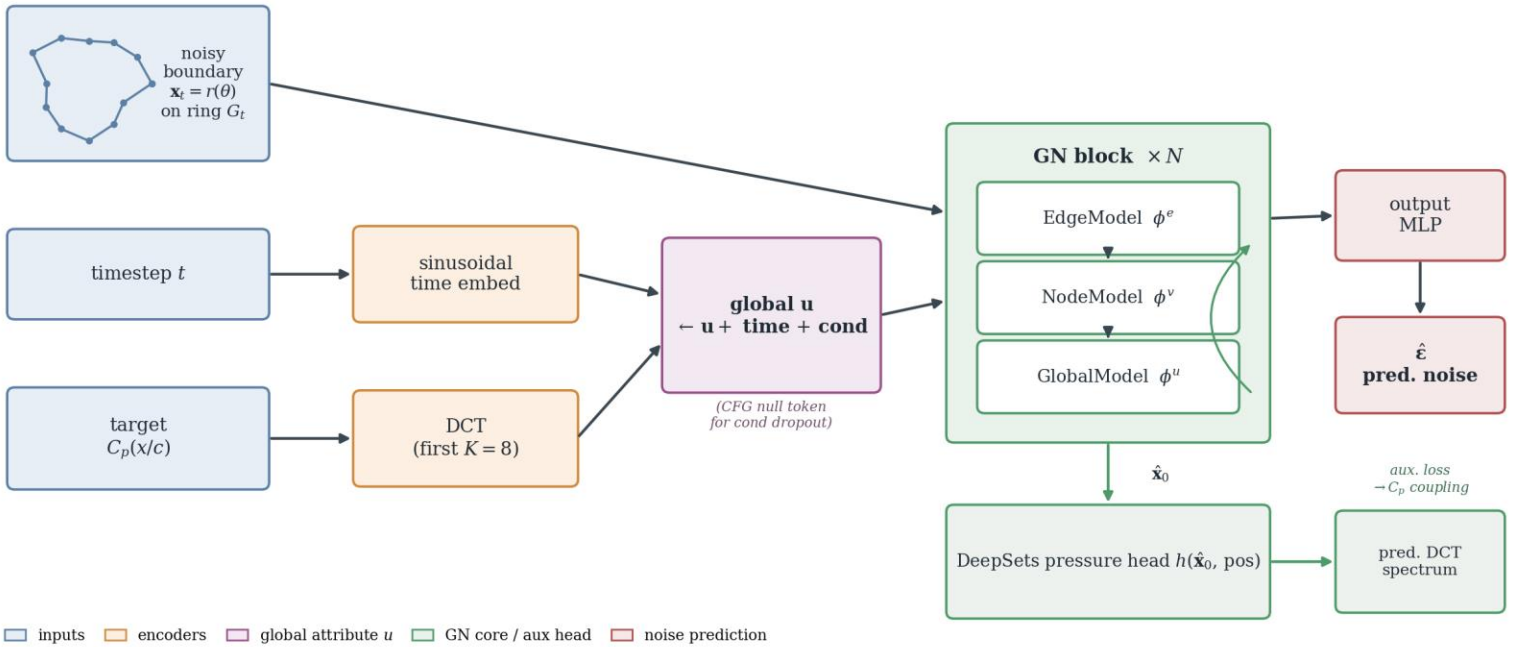
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Extensions to this Work

- Improve Model
  - Add new conditioning parameters
  - Change geometry representations
- Gradient Free Optimisation (Low Fidelity)
- Relevant Applications
  - Cycling geometries (helmet, fork)
  - Aero shape optimisation
  - Cardiac Electrophysiology
  - Anything with a surface

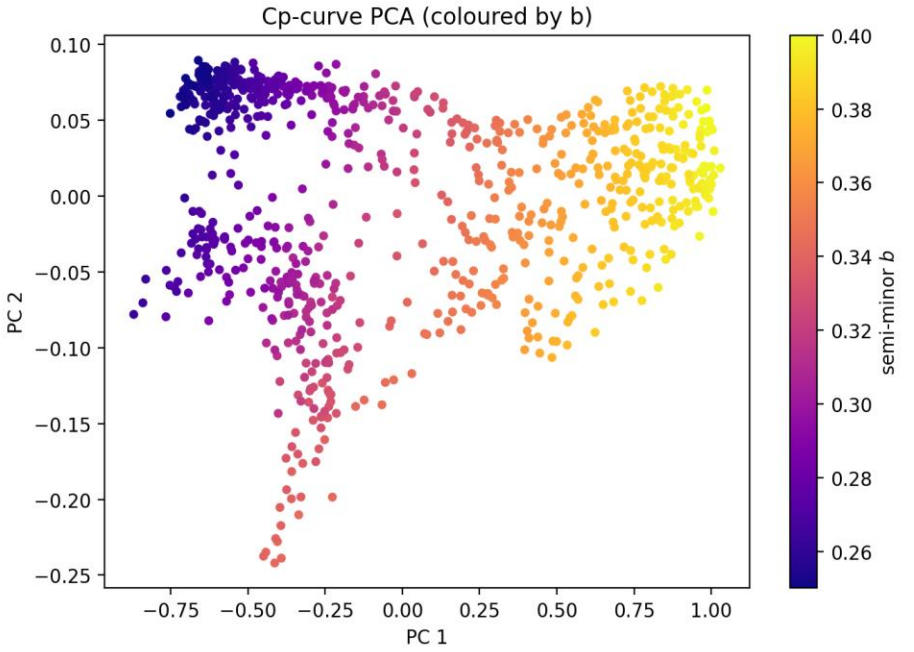
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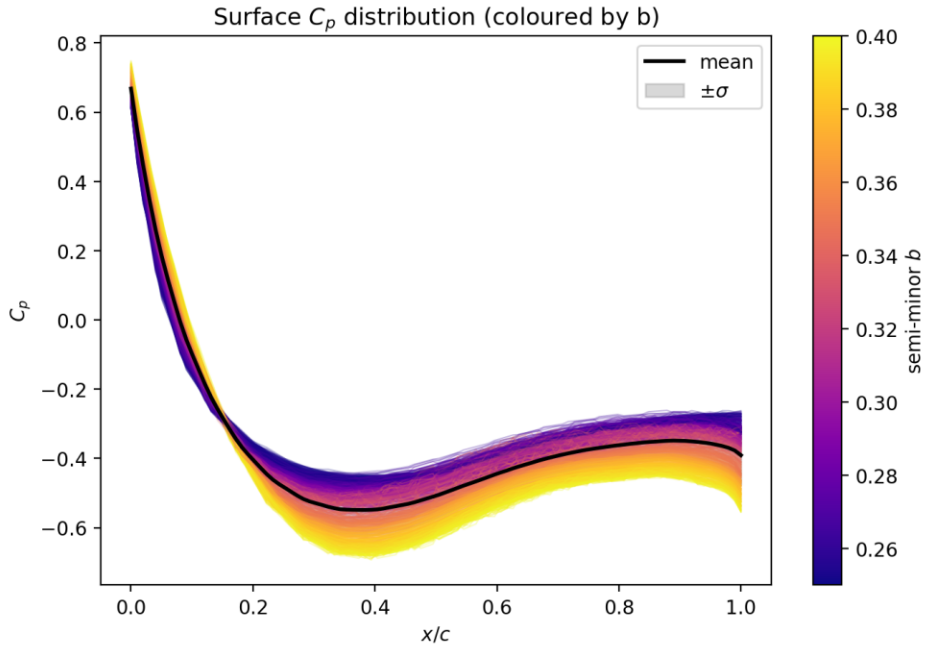
Thank you --- Questions

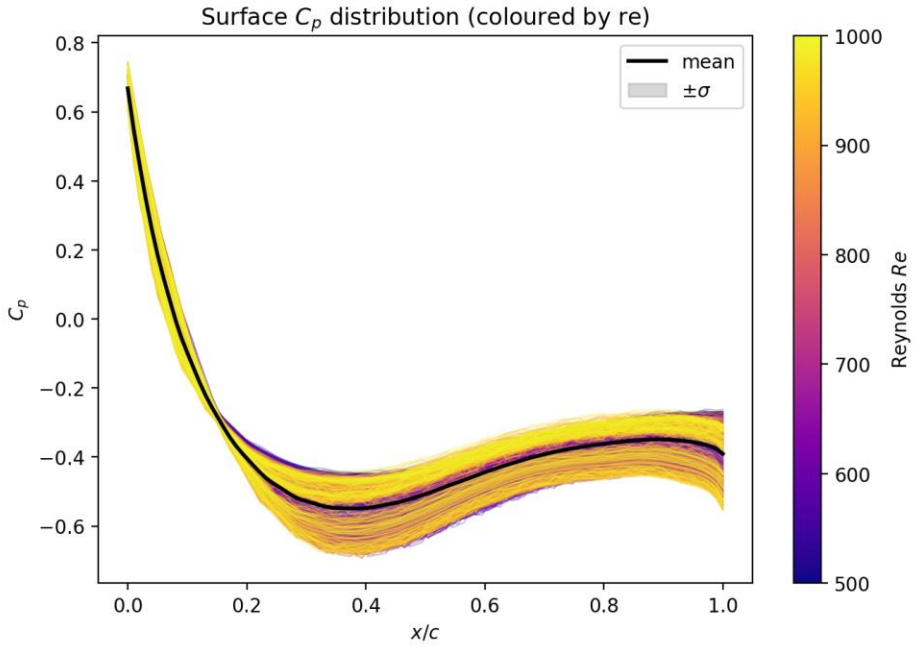


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Appendix B - Dataset

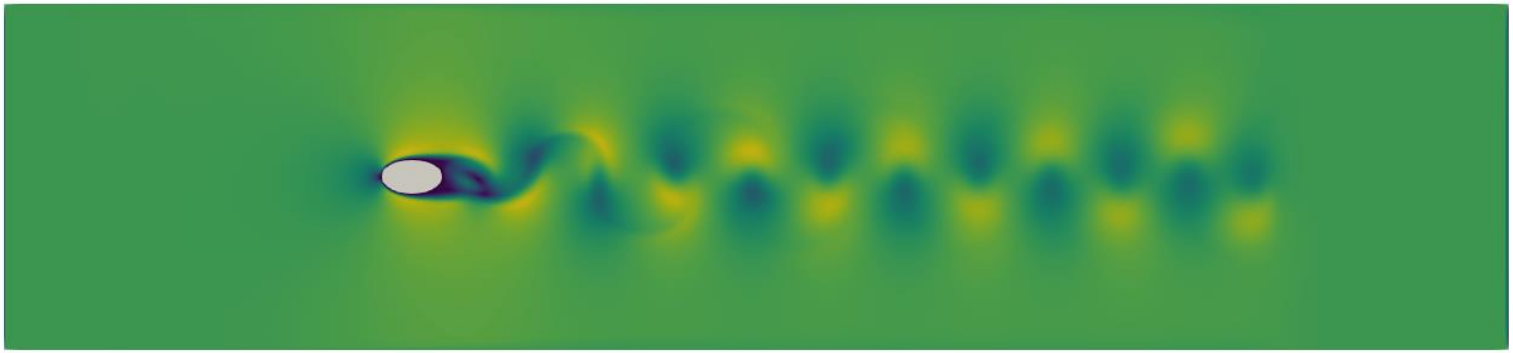




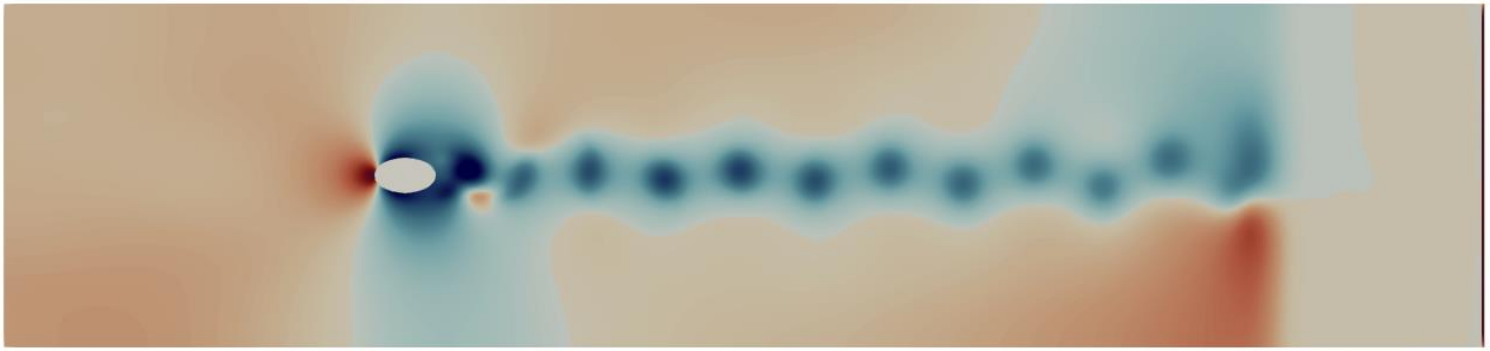


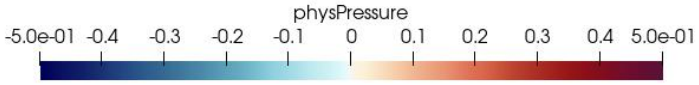
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Appendix C - 2D LBM Flow Fields









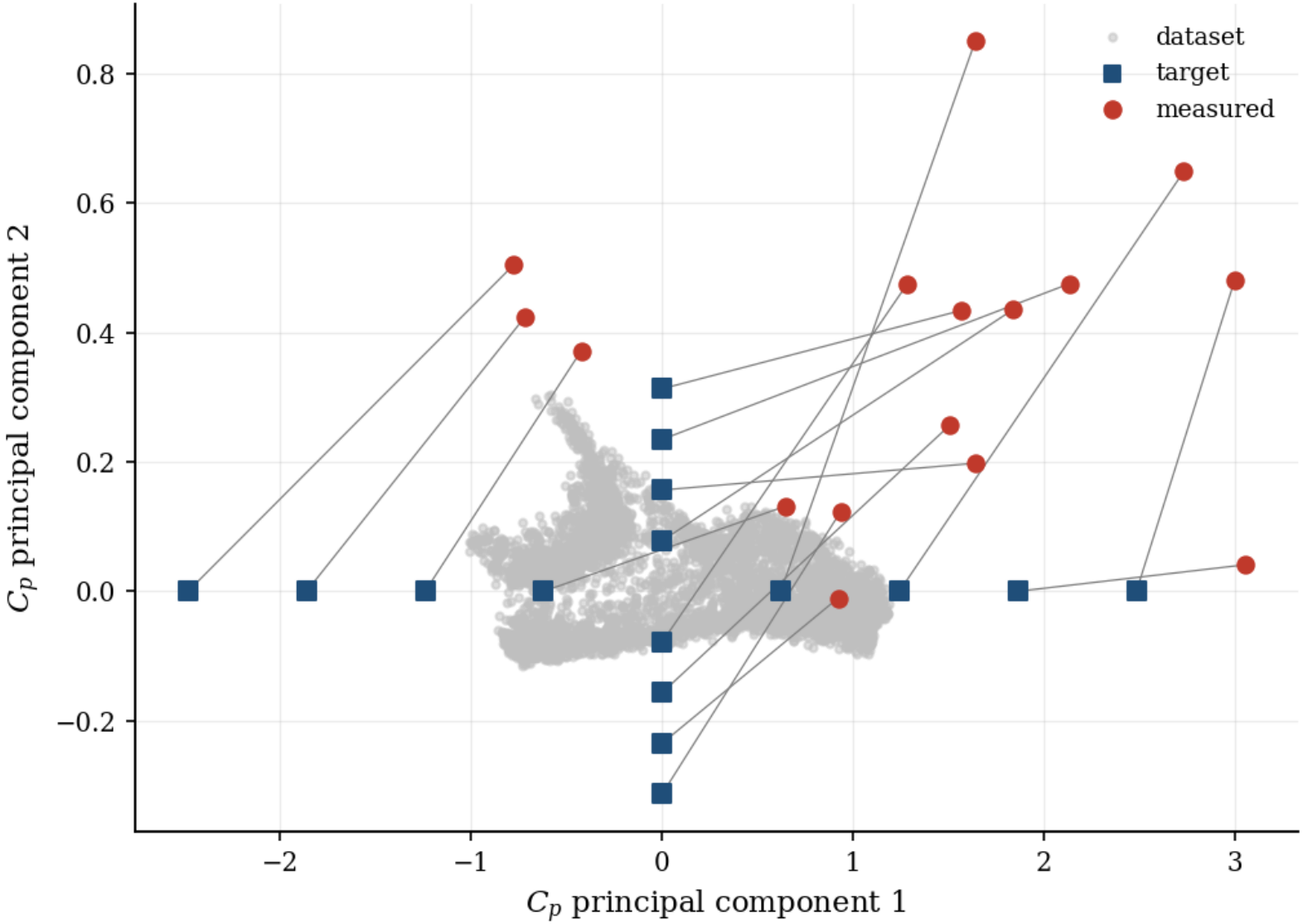
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LBM Setup

Quantity	Value (as run)
Method	D2Q9, single-relaxation BGK
Reynolds number	500 – 1000
Mach number	≈ 0.13 (held fixed)
Relaxation τ / ω	0.545 / 1.835
Resolution	N ≈ 0.2·Re cells/chord (100 → 200)
Knudsen number	$\approx 3 \times 10^{-4}$
Wall model	Bouzidi curved no-slip
Turbulence model	none (DNS-resolved)
Domain	24 chords (6·body·17) + sponge outlet
Time-averaging	≥ 10 shedding periods after washout
Cost (1 GPU, ≈ 1500 MLUPs)	60 s (Re500) → 7.5 min (Re1000)

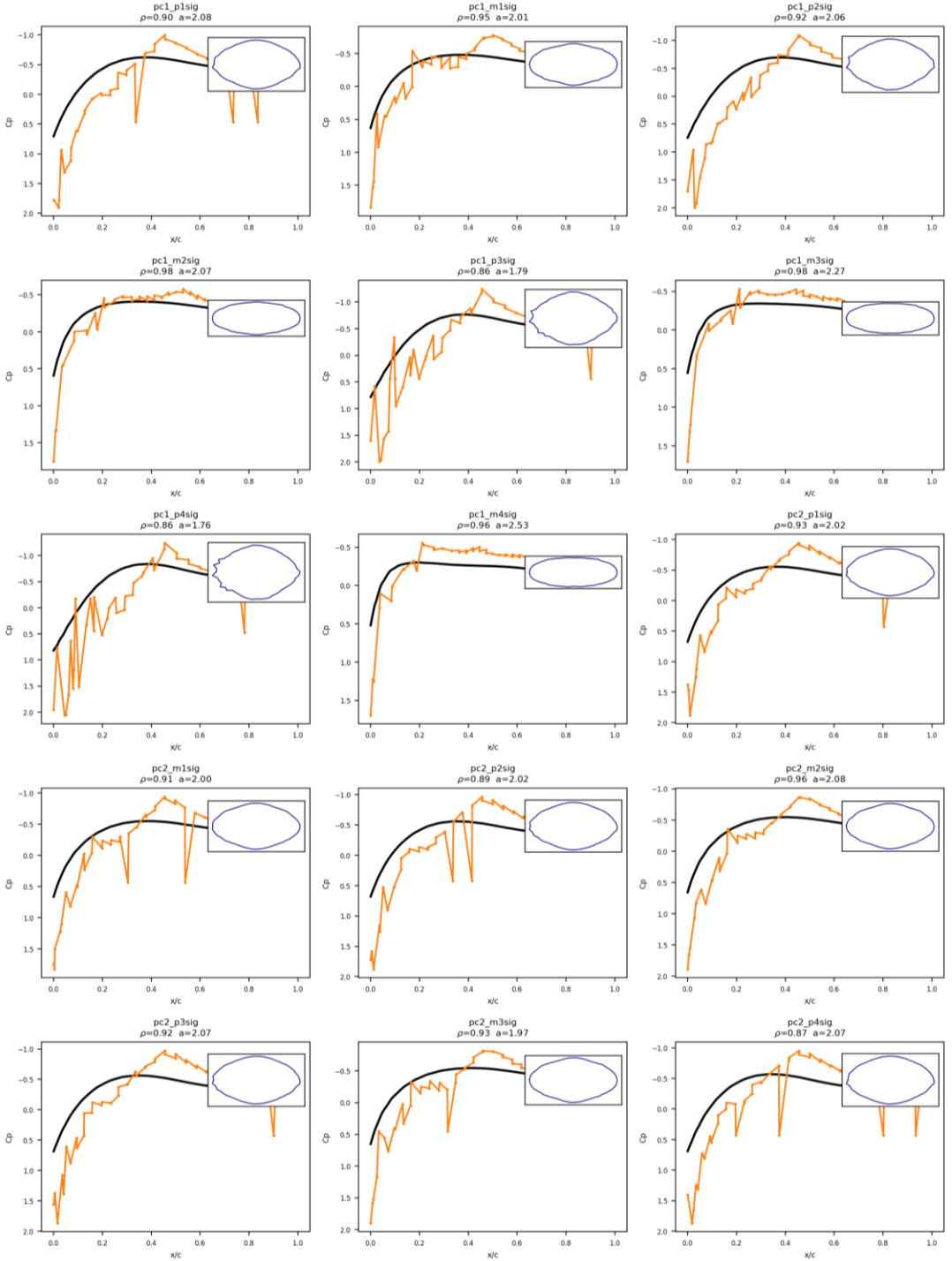
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Appendix D – Out of Distribution



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